

Aim

24000329a1

Q: Students are required to design a small embedded system application of your choice (for example: Smart temperature logger, motor controller, IoT sensor node, digital attendance system, traffic light controller, wearable device, etc..)

to implement your applications, you must search, read and analyze real microcontroller datasheets from any semiconductor vendor (such as ST microelectronics, ARM, TI, Arduino etc.) and select the most suitable MCU

A. Embedded system Design - Digital attendance System purpose:

A Digital Attendance System automatically records attendance using RFID cards as biometric input reducing manual effort, proxy attendance and errors.

It is commonly used in colleges, offices and organizations functional requirements.

- identify user using RFID card or fingerprint
- verify user ID and with stored data
- Record date and time of attendance
- Display attendance status
- store attendance data locally or upload to server / cloud

Requirements :

Hardware Requirements

- RFID Reader or PingPong
- RFID Cards
- microcontroller Unit
- LCD / OLED
- Buzzer
- Power supply

selected MCU: Arduino Uno (Atmega 328P)

MCU Datasheet - Based analysis

- 8bit AVR microcontroller
- 16MHz clock frequency
- 32KB flash memory
- 2KB SRAM 1KB EEPROM
- JPI, I2C, UART communication
- low power consumption

Justification for MCU selection

- Sufficient memory and speed for attendance log
- Easy interfacing with RFID and Biometric modules
- Built-in EEPROM useful for storing user IDs
- Simple programming and strong community support
- Cost effective and ideal for academic projects

Software Requirements

- Embedded C / C++
- MCU Board support Package
- RFID Biometric Libraries
- EEPROM / SPI
- Serial communication

Simple user

1. user pin

2. Reader

3. MCU

4. Attendance

5. status

Block

RFID
Fingerprint

Con

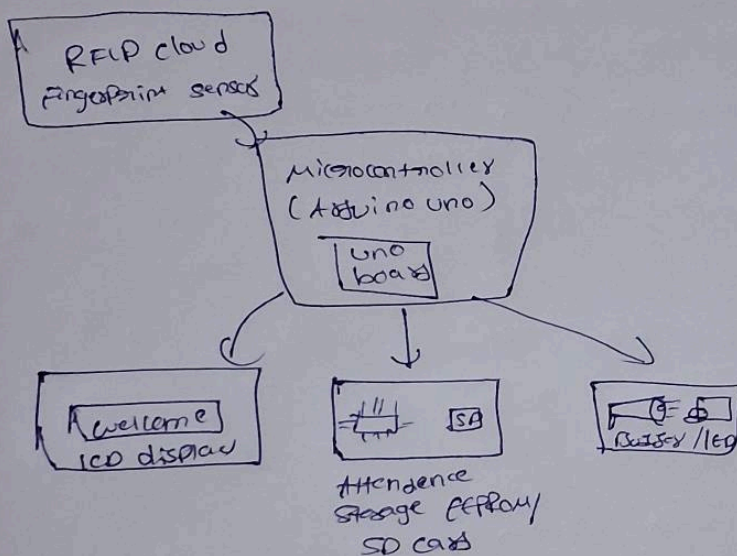
-

a

Simple working principle

1. user places RFID card or finger on sensor
2. Reader sends ID data to Arduino
3. MCU verifies ID with stored records
4. Attendance is logged with date and time
5. status is displayed and LED indicates success

Block diagram



Conclusion :

The Arduino uno-based attendance system is a simple, reliable and efficient embedded solution based on the Atmega 328 data

Sheet

9/1/26
2400032991